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T11 - 05

Assignment No. 8

Aim: Add a service worker to PWA for caching and fetching web resource

Theory:

Working of Cache API with Service Workers in Web Applications:

What is a Service Worker?

A Service Worker is a JavaScript file that runs in the background of a web application, separate from the main browser thread. It is designed to handle tasks like:

- Caching resources
- Fetching data from the cache when offline
- Managing network requests

What is the Cache API?

The Cache API is a web storage API used by service workers to store network responses. These cached responses can later be retrieved and served when the network is unavailable or slow, allowing web apps to work offline or load faster.

How It Works

1. Service Worker Registration

The service worker is registered in the main JavaScript file of your app (e.g., index.js).

2. Install Event

During the install event, static files (HTML, CSS, JS, images, etc.) are cached using the Cache API.

3. Fetch Event

The service worker intercepts every network request and checks if the requested file is available in the cache. If yes, it serves it from the cache. If not, it fetches from the network.

4. Offline Access

If the user is offline, cached resources are served, ensuring the app still works partially or fully.

Sample Service Worker Script:

```
serviceWorker.js const CACHE_NAME = "car-
rental-cache-v1"; const urlsToCache = [
  "/",
  "/index.html",
  "/styles.css",
  "/app.js",
  "/icon192.png",
  "/icon144.png"
];

// Install event – caching resources
self.addEventListener("install", (event) => {
  event.waitUntil(
```

```

caches.open(CACHE_NAME).then((cache) => {
  return cache.addAll(urlsToCache);

})

);

});

// Fetch event – serving from cache if available
self.addEventListener("fetch", (event) => {
  event.respondWith(
    caches.match(event.request).then((response) => {

      // Return cached response if
      available if (response) { return
      response;

    }

    // Otherwise fetch from network
    return fetch(event.request);

  })

});

```

How to Register the Service Worker (in

index.js) if ("serviceWorker" in navigator) {

```
window.addEventListener("load", () => {  
  navigator.serviceWorker  
    .register("/serviceWorker.js")  
  .then((registration) => {  
    console.log("ServiceWorker registered: ",  
      registration);  
  })  
  .catch((error) => {    console.log("ServiceWorker  
registration failed: ", error);  
  });  
});  
}
```

Benefits of Using Cache API with Service Workers

- Enables offline functionality
- Improves load time by serving cached content
- Reduces server load and bandwidth
- Provides a better user experience

Output:

Save Rental

big

with our car rental

Rent the car of your dreams. Unbeatable prices, unlimited miles, flexible pick-up options and much more.

Book Ride

Application

- Manifest
- Service workers
- Storage

Storage

- Local storage
- Session storage
- Extension storage
- IndexedDB
- Cookies
- Private state tokens
- Interest groups
- Shared storage
- Cache storage
 - car-rental-cache-v1...
- Storage buckets

Background services

- Back/forward cache
- Background fetch
- Background sync
- Bounce tracking miti...
- Notifications
- Payment handler
- Periodic background ...
- Speculative loads

Service workers

Offline

Update on reload

Bypass for network

http://localhost:3000/

Source [sw.js](#)

Received 4/17/2025, 5:37:47 PM

Status ● #4299 activated and is running

Stop

Push

Test push message from DevTools..

Push

Sync

test-tag-from-devtools

Sync

Periodic sync

test-tag-from-devtools

Periodic sync

Update Cycle

Version

Update Activity

Timeline

#4299

Install

#4299

Wait

#4299

Activate

Service workers from other origins

[See all registrations](#)

Service workers

Filter by path

http://localhost:3000

Origin http://localhost:3000

Bucket name default

Is persistent No

Durability relaxed

Quota 0 B

Expiration None

#	Name	Response-Type	Content-Type	Content-Length	Time Cached	Vary Header
0	/	basic	text/html	0	4/17/2025, 5:3...	Accept-Encodi...
1	/icon144.png	basic	image/png	3,686	4/17/2025, 5:3...	
2	/icon192.png	basic	image/png	5,367	4/17/2025, 5:3...	
3	/index.html	basic	text/html	0	4/17/2025, 5:3...	Accept-Encodi...

Total entries: 4

